ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

Model Answers to Tutorial Questions

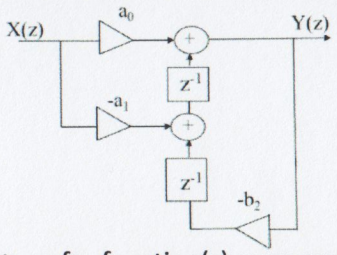
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Term 3, 2021

Topic 3: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

1. A digital filter structure is shown below. Calculate the dc gain of the filter, in terms of the filter coefficients. (Tick the correct answer)



(a) $1 + a_0 - a_1$

✓ (b) $\frac{a_0 - a_1}{1 + b_2}$

(c) $\frac{a_0 - a_1}{1 - b_2}$

(d) Non of the above

[see Page 13, Topic 3]

$H(z) = \frac{a_0 + a_1 z^{-1}}{1 + b_2 z^{-2}}$, $\therefore H(\theta) = \frac{a_0 - a_1 e^{-j\theta}}{1 + b_2 e^{-j2\theta}}$

DC gain $= H(\theta)\Big|_{\theta=0} = \frac{a_0 - a_1}{1 + b_2}$

2. Tick which of the following transfer function(s) represent high pass filters, where $0 < \alpha < 1$.

✓ (a) $H(z) = 1 - \alpha z^{-1}$ (b) $H(z) = \frac{1}{1 - \alpha z^{-1}}$

(c) $H(z) = 1 + \alpha z^{-1}$ ✓ (d) $H(z) = \frac{1}{1 + \alpha z^{-1}}$

(b) & (c) are Lowpass filters (Topic 2: Page 34)

See Example 17: Topic 2 (Page 35)
(a) & (d) are Highpass filters

3. The Digital Filter $H(z) = \frac{2 + 2z^{-1}}{3 - z^{-1}}$; (Tick the correct answer)

✓ a) is a low-pass filter
b) is a high-pass filter
c) is an all-pass filter

$H(z) = \frac{2}{3} \frac{(1 + z^{-1})}{1 - \frac{1}{3} z^{-1}}$

$1 + z^{-1}$ is a Lowpass filter
& $\frac{1}{1 - \alpha z^{-1}}$ is a Lowpass filter
(Topic 2: Page 34)

4. A maximum phase filter has: (Tick the correct answer)

a) All zeros and poles outside the unit circles
b) All poles outside the unit circle
✓ c) All zeros outside the unit circle
d) All poles and zeros inside the unit circle

See Page 32, Topic 3

Topic 3: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

5. Which of the following is/are linear phase filters: (Tick the correct answer)

✓(a) $H(z) = 0.25 + 0.5\,z^{-1} + 0.25\,z^{-2}$ ✓(b) $H(z) = z^{-4}$

✓(c) $H(z) = 1 - z^{-2}$ (d) $H(z) = 1 + 2z^{-1} + 2\,z^{-2} - z^{-3}$

(b) See Topic 3, Page 26
(a) impulse response are symmetric about
(c) its mid Point (Page 8, Topic 3)

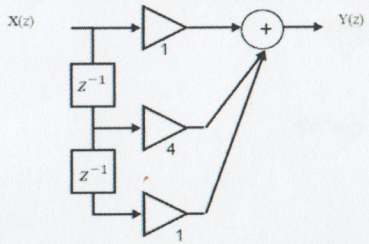
6. What can be said of the following filter ? (Tick the correct answer)

a) It has minimum phase and a finite impulse response.

b) It has linear phase and an infinite impulse response.

c) It has an infinite impulse response.

✓d) It has linear phase and a finite impulse response



$* H(z) = 1 + 4z^{-1} + z^{-2} = \frac{z^2 + 4z + 1}{z^2}$

$z = \frac{-4 \pm \sqrt{16-4}}{2}$

$z = -2 \pm \sqrt{3}$

one zero inside the unit circle & one outside the unit circle

(a) is wrong

* FIR filter
∴ (b) & (c) are wrong

* Symmetrical filter coefficients — linear Phase

7. Which of the following system is linear, time invariant, causal and stable? (Tick the correct answer)

a) y[n] = |x[n]| ← non linear

b) y[n] = x[n] + n x[n+1] ← Time Variant

c) y[n] = x[-n] ← non causal

✓ *d)* $y[n] = \frac{1}{N}\sum_{k=0}^{N-1} x[n-k]$ (an FIR filter)

Topic 3: Tutorial Questions (Level 2)

SOLO – Level 2

1. Compute the 3dB bandwidth of the following filters:

$$H_1(z) = \frac{1-a}{1-az^{-1}};\quad H_2(z) = \frac{1-a}{2}\frac{1+z^{-1}}{1-az^{-1}},\qquad 0 < a < 1$$

Which filter has a smaller 3dB bandwidth?

$H_1(\theta) = \frac{1-a}{1-ae^{-j\theta}}$; $|H_1(\theta)|^2 = H_1(\theta)\cdot H_1^*(\theta)$

$= \frac{1-a}{1-ae^{-j\theta}}\cdot\frac{1+a}{1-ae^{j\theta}}$

$= \frac{(1-a)^2}{1+a^2-2a\cos\theta}$

$|H_1(\theta)|^2_{\theta=\theta_{c1}} = \frac{1}{2}$

$\frac{(1-a)^2}{1+a^2-2a\cos\theta_{c1}} = \frac{1}{2} \Rightarrow \cos\theta_{c1} = \frac{4a-a^2-1}{2a}$

$H_2(\theta) = \frac{1-a}{2}\frac{1+e^{-j\theta}}{1-ae^{-j\theta}}$

$|H_2(\theta)|^2 = \left(\frac{1-a}{2}\right)^2 \frac{(1+\cos\theta)^2+\sin^2\theta}{(1-a\cos\theta)^2+a^2\sin^2\theta} = \left(\frac{1-a}{2}\right)^2 \frac{2+2\cos\theta}{1+a^2-2a\cos\theta}$

$|H_2(\theta)|^2_{\theta=\theta_{c2}} = \frac{1}{2} \Rightarrow \frac{1}{2} = \left(\frac{1-a}{2}\right)^2 \frac{2+2\cos\theta_{c2}}{1+a^2-2a\cos\theta_{c2}}$

$\therefore \boxed{\cos\theta_{c2} = \frac{2a}{1+a^2}}$

$\cos\theta_{c2} > \cos\theta_{c1}$ for $0<a<1$

$\therefore \theta_{c2} < \theta_{c1}$

2. Find the magnitude response of the filter *H(z)* and comment on your result.

$$H(z) = \frac{1+3z^2}{3+z^2}$$

Draw the canonic realization of the filter *H(z)*.

$H(\theta) = \frac{1+3e^{j2\theta}}{3+e^{j2\theta}}$ $\quad |H(\theta)|^2 = H(\theta)\cdot H^*(\theta)$

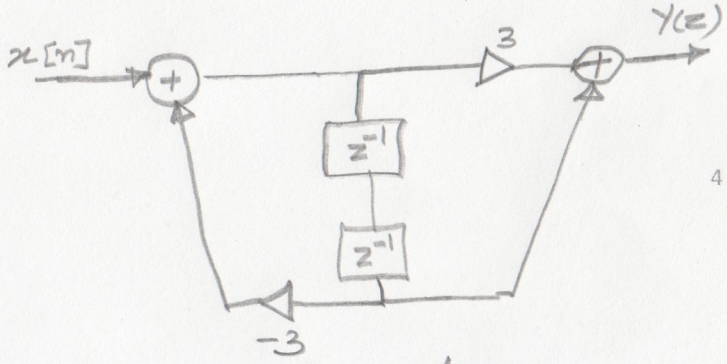
$|H(\theta)|^2 = \left(\frac{1+3e^{j2\theta}}{3+e^{j2\theta}}\right)\left(\frac{1+3e^{-j2\theta}}{3+e^{-j2\theta}}\right)$

$= \frac{(1+3\cos2\theta)^2+(3\sin2\theta)^2}{(3+\cos2\theta)^2+(\sin2\theta)^2} = \frac{10+6\cos2\theta}{10+6\cos2\theta}$

$= 1$

$\boxed{|H(\theta)| = 1}$

$H(z) = \frac{1+3z^2}{3+z^2} = \frac{3+z^{-2}}{1+3z^{-2}}$

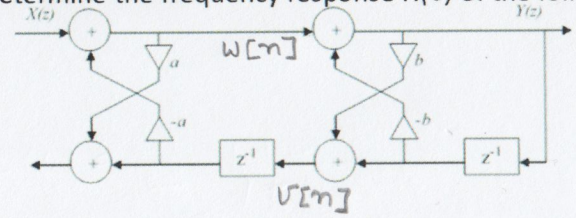


canonic form.

Topic 3: Tutorial Questions (Level 2)

SOLO – Level 2

3. Determine the frequency response $H(\theta)$ of the following filter:



$y[n] = w[n] - by[n-1]$ — ①
$v[n] = y[n-1] + by[n]$ — ②
$w[n] = x[n] - av[n-1]$ — ③

Substitute ③ in ① to eliminate $w[n]$

$\therefore y[n] = x[n] - av[n-1] - by[n-1]$

$v[n-1] = y[n-2] + by[n-1]$

$y[n] = x[n] - ay[n-2] - ab\,y[n-1] - by[n-1]$

$\frac{Y(z)}{X(z)} = \frac{1}{1 + b(a+1)z^{-1} + az^{-2}}$

$\therefore H(\theta) = \frac{1}{1 + b(a+1)e^{-j\theta} + ae^{-j2\theta}}$

4. A two-pole lowpass filter has the transfer function

$$H(z) = \frac{k_0}{(1 - b_1 z^{-1})^2}$$

Determine the values of k_0 and b_1 such that the frequency response $H(\theta)$ satisfy the conditions

$$H(0) = 1; \quad |H(\theta)|^2_{\theta=\frac{\pi}{4}} = \frac{1}{2}$$

$H(\theta) = \frac{k_0}{(1 - b_1 e^{-j\theta})^2}$ · $|H(0)| = 1 \Rightarrow 1 = \frac{k_0}{(1-b_1)^2}$

$\therefore k_0 = (1-b_1)^2$

$|H(\theta)|^2 = \frac{k_0}{(1-b_1 e^{j\theta})^2} \cdot \frac{k_0}{(1-b e^{j\theta})^2} = k_0^2 \frac{1}{(1+b_1^2 - 2b_1\cos\theta)^2}$

$|H(\theta)|^2_{\theta=\pi/4} = \frac{1}{2} = \frac{k_0^2}{\left(1+b_1^2 - 2b_1\frac{1}{\sqrt{2}}\right)^2} \Rightarrow (1+b_1^2 - b_1\sqrt{2})^2 = 2k_0^2$

$\therefore (1+b_1^2 - b_1\sqrt{2}) = \sqrt{2}(1-b_1)$

$\therefore b_1 = 0.32$
& $k_0 = (1-0.32)^2 = 0.46$

5. Implement a parallel structure for the following $H(z)$:

$$H(z) = \frac{z^2 - 4z + 3}{z(z+0.5)^2}$$

$\frac{z^2-4z+3}{z(z+0.5)^2} = \frac{A}{z} + \frac{Bz+C}{(z^2+z+\frac{1}{4})}$

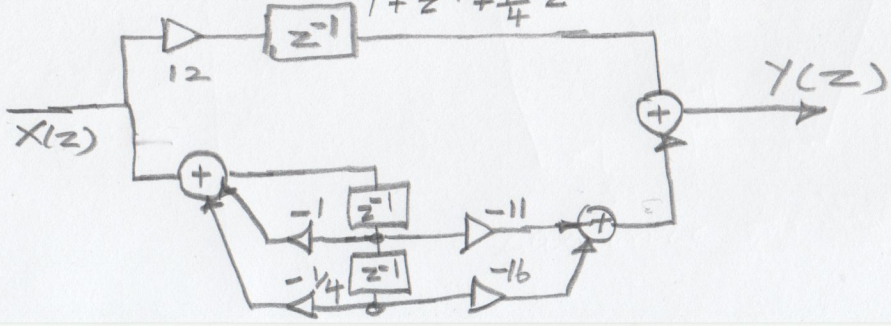
$\therefore z^2 - 4z + 3 = Az^2 + Az + \frac{A}{4} + Bz^2 + Cz$

By equating the coefficients, we obtain

$A+B=1$, $A+C=-4$, $\frac{A}{4} = 3$

$\therefore B = -11$, $C = -16$ & $A = 12$

$H(z) = 12 + \frac{-11z^{-1} - 16z^{-2}}{1 + z^{-1} + \frac{1}{4}z^{-2}}$



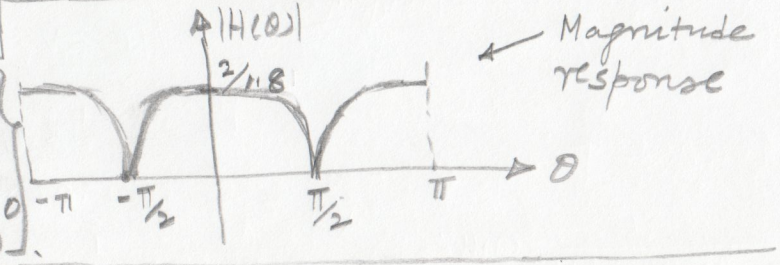
Topic 3: Tutorial Questions (Level 2)

SOLO – Level 2

6. A notch filter is given by $H(z)=\frac{1+z^{-2}}{1+0.8z^{-2}}$. Determine the magnitude response at $\theta=0,\frac{\pi}{2}$, *and* π. Sketch the magnitude response in the interval $-\pi\le\theta\le\pi$.

$H(\theta)=\frac{1+e^{-j2\theta}}{1+0.8e^{-j2\theta}}$ at $\theta=\pi/2$ $\quad H(\theta)\big|_{\theta=\pi/2}=\frac{1+e^{-j\frac{\pi}{2}\times2}}{1+0.8e^{-j\frac{\pi}{2}\times2}}=\frac{1+\cos\pi-j\sin\pi}{1+0.8(\cos\pi-j\sin\pi)}=0$

DC gain $H(\theta)\big|_{\theta=0}=\frac{2}{1.8}$ $\quad$ at $\theta=\pi$, $H(\theta)\big|_{\theta=\pi}=\frac{1+e^{-j2\pi}}{1+0.8(\cos\pi-j\sin\pi)}=\frac{2}{1.8}$



Magnitude response

7. Determine and sketch the magnitude and phase response of the following filters

(a) $y[n]=\frac{1}{2}(x[n]-x[n-1])$ (b) $y[n]=x[n]-x[n-8]$

(c) $y[n]=x[n-4]$

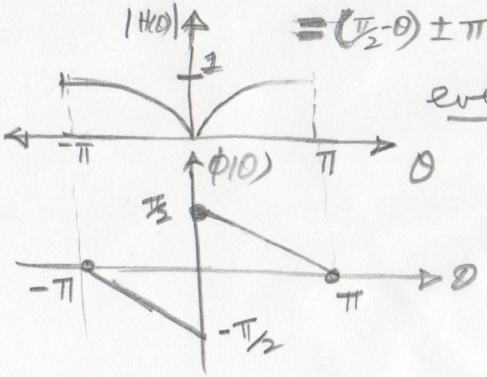
(a) $Y(z)=\frac{1}{2}X(z)-\frac{1}{2}z^{-1}X(z)\Rightarrow H(z)=\frac{1}{2}(1-z^{-1})$

$H(\theta)=\frac{1}{2}(1-e^{-j\theta})=\frac{1}{2}e^{-j\frac{\theta}{2}}\frac{e^{j\frac{\theta}{2}}-e^{-j\frac{\theta}{2}}}{(2j)}(2j)$

$=je^{-j\frac{\theta}{2}}\sin\frac{\theta}{2}=e^{j\frac{\pi}{2}}e^{-j\frac{\theta}{2}}\sin\frac{\theta}{2}=e^{j(\frac{\pi}{2}-\theta)}\cdot\underbrace{\sin\frac{\theta}{2}}_{B(\theta)}$

$|H(\theta)|=|\sin\frac{\theta}{2}|$; $\phi(\theta)=(\frac{\pi}{2}-\theta)$ 当 $B(\theta)\ge0$

$=(\frac{\pi}{2}-\theta)\pm\pi$ $B(\theta)<0$



even function

odd function

$y[n]=x[n]-x[n-8]$

$H(z)=\frac{Y(z)}{X(z)}=1-z^{-8}$

$H(\theta)=1-e^{-j8\theta}=e^{-j4\theta}[e^{+j4\theta}-e^{-j4\theta}]$

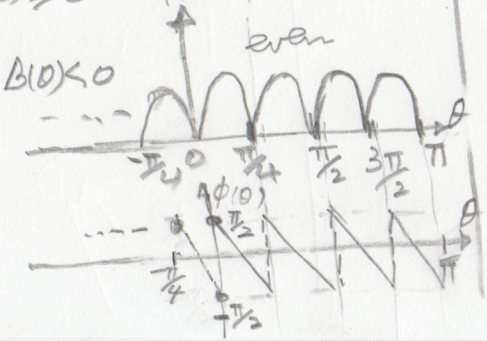
$H(\theta)=e^{-j4\theta}(2j)\sin4\theta$

$H(\theta)=e^{j(\frac{\pi}{2}-4\theta)}\underbrace{2\sin4\theta}_{=B(\theta)}$

$|H(\theta)|=|2\sin4\theta|$

$\phi(\theta)=\frac{\pi}{2}-4\theta$ $B(\theta)\ge0$

$=(\frac{\pi}{2}-4\theta)\pm\pi$ $B(\theta)<0$

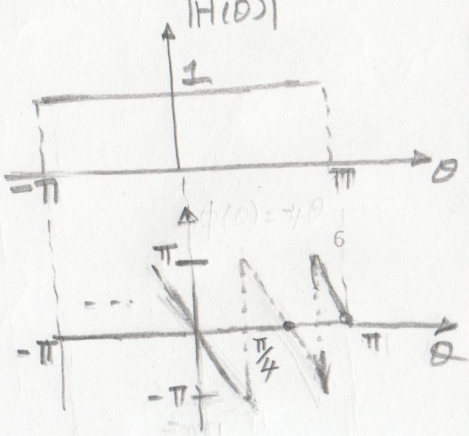


(c) $Y(z)=z^{-4}X(z)$

$H(z)=z^{-4}$; $H(\theta)=e^{-j4\theta}$

$|H(\theta)|=1$

$\phi(\theta)=-4\theta$

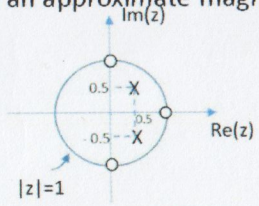


Topic 3: Tutorial Questions (Level 2)

SOLO – Level 2

8 (a) Determine the transfer function H(z) of a discrete - time system with the pole-zero map shown below:

(b) Using the pole-zero map , sketch an approximate magnitude response of the H(z) .



radius of the pole $(r) = \sqrt{(0.5)^2+(0.5)^2} = \frac{1}{\sqrt{2}}$

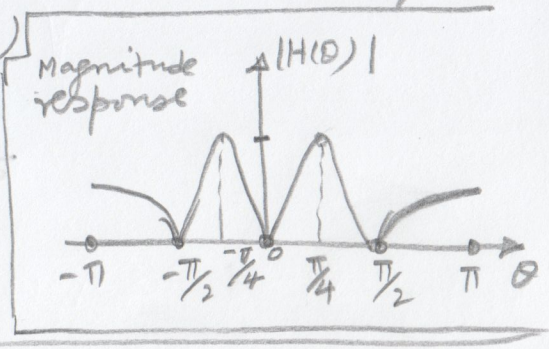
angle of the pole $= \theta$; $\tan\theta = 0.5/0.5 = 1$

$\theta = \pi/4$

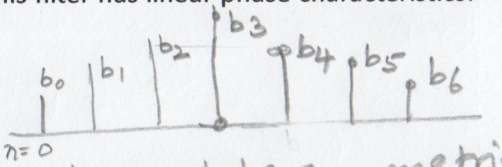
$$H(z) = \frac{(z-1)(z+j)(z-j)}{(z-re^{-j\frac{\pi}{4}})(z-re^{j\frac{\pi}{4}})}$$

$$= \frac{(z-1)(z^2+1)}{z^2 - \frac{2}{\sqrt{2}}\cos\frac{\pi}{4}z + \frac{1}{2}}$$

$$H(z) = \frac{(z-1)(z^2+1)}{z^2 - z + \frac{1}{2}}$$



9. A Finite impulse filter filter has seven filter coefficients b_0 to b_6, with $b_0 = 1$, $b_1 = 2$, $b_2 = 3$ and $b_3 = 4$.Determine b_4 to b_6 such t hat this filter has linear phase characteristics.



The coefficients must be symmetrical about b_3 for a linear phase filter $\Rightarrow$

$b_4 = b_2 = 3$ ✓

$b_5 = b_1 = 2$

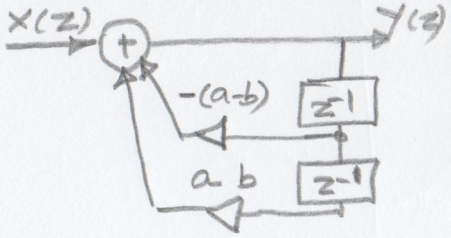
$b_6 = b_0 = 1$

10. Implement the following transfer function in cascade structure , in direct form II and in parallel structure. All coefficients are real.

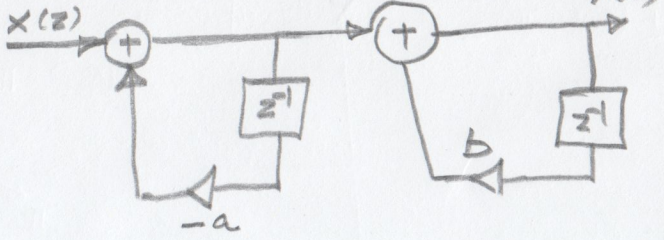
$$H(z) = \frac{1}{(1+az^{-1})(1-bz^{-1})}$$

Direct form II

$$H(z) = \frac{1}{1+(a-b)z^{-1}-abz^{-2}}$$



Cascade

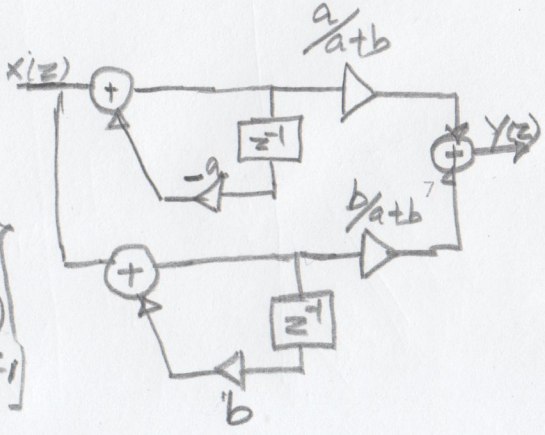


Parallel structure

$$H(z) = \frac{1}{(1+az^{-1})(1-bz^{-1})}$$

$$= \frac{A}{1+az^{-1}} + \frac{B}{1-bz^{-1}}$$

$$H(z) = \frac{(a/a+b)}{1+az^{-1}} + \frac{(b/a+b)}{1-bz^{-1}}$$



Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

1. Show that both digital filters given below have the same magnitude response:

   (a) $y[n] = \sum_{i=-m}^{m} c_i\, x[n-i]$ (b) $y[n] = \sum_{i=-m}^{m} c_i\, x[n-m-i]$

   where $y[n]$ is the output, $x[n]$ is the input and c_i are the coefficients.

$$Y(z) = \sum_{i=-m}^{m} c_i\, X(z)\cdot z^{-i}$$

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{i=-m}^{m} c_i\, z^{-i} \Rightarrow \boxed{|H_1(\theta)| = \left|\sum_{i=-m}^{m} c_i e^{-j\theta i}\right|}$$

$$Y(z) = \sum_{i=-m}^{m} c_i\, X(z)\cdot z^{-m}\cdot z^{-i}$$

$$H_2(z) = \frac{Y(z)}{X(z)} = \sum_{i=-m}^{m} c_i\, z^{-m}\cdot z^{-i}$$

$$H_2(\theta) = \sum_{i=-m}^{m} c_i\, e^{-j\theta i}\cdot e^{-j\theta m}$$

$$\boxed{|H_2(\theta)| = \left|\sum_{i=-m}^{m} c_i\, e^{-j\theta i}\right|\cdot\underbrace{\left|e^{-j\theta m}\right|}_{=1}}$$

$$\therefore \underline{|H_1(\theta)| = |H_2(\theta)|}$$

2. Consider the following causal IIR transfer function:

$$H(z) = \frac{2z^3 - 4z^2 + 9}{(z-3)(z^2+z+0.5)}$$

Is *H(z)* a stable function? If it is not stable, find a stable transfer function *G(z)* such that $|G(\theta)| = |H(\theta)|$. Is there any transfer function having the same magnitude response as *H(z)*?

Since H(z) has a pole at z=3 (outside the unit circle), the given transfer function is unstable. To construct a stable transfer function having the same magnitude response consider another transfer function:

$$G(z) = H(z)\cdot A(z) \quad \leftarrow \text{Make } A(z) \text{ as an all-pass filter}$$

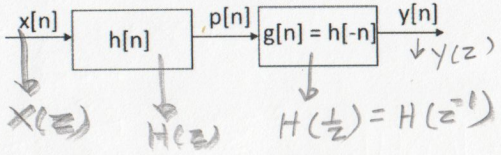
$$= \frac{2z^3-4z^2+9}{(z^2+z+0.5)(z-3)}\cdot\underbrace{\frac{z-3}{1-3z}}_{\text{all-pass filter (Magnitude response is constant or 1)}}$$

$$\therefore |G(\theta)| = |H(\theta)||A(\theta)| \Rightarrow |G(\theta)| = |H(\theta)|$$

Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

3. Show that if $x[n]$ is a real valued signal, the structure shown below has a zero-phase response; where $h[n]$ is the impulse response of the filter H(z) given by; $H(z) = (\frac{1}{2} + z^{-1} + \frac{1}{2}z^{-2})$.



4. The difference equation of a digital filter is given by

$$y[n] = (1/8)\{x[n] - x(n-8)\} - y(n-2)$$

(a) Find the transfer function for the above filter.

(b) Using the transfer function obtained in part (a) above, draw the pole zero map for the filter and sketch the corresponding magnitude response.

$\frac{Y(z)}{X(z)} = H(z) \cdot H(z^{-1})$ or $\frac{Y(\theta)}{X(\theta)} = H(\theta) \cdot H^*(\theta)$

$H(\theta) = \frac{1}{2} + e^{-j\theta} + \frac{1}{2}e^{-j2\theta}$; $H^*(\theta) = \frac{1}{2} + e^{j\theta} + \frac{1}{2}e^{j2\theta}$

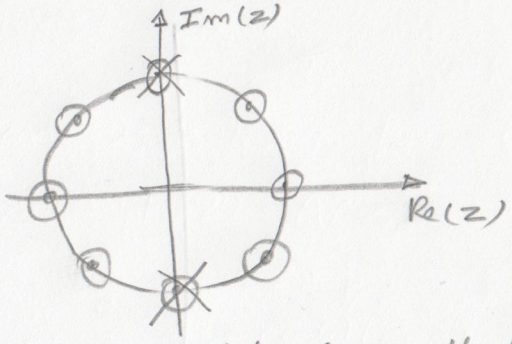
$\frac{Y(\theta)}{X(\theta)} = \left(\frac{1}{2} + e^{-j\theta} + \frac{1}{2}e^{-j2\theta}\right)\left(\frac{1}{2} + e^{j\theta} + \frac{1}{2}e^{j2\theta}\right)$

$= \frac{3}{2} + (e^{j\theta} + e^{-j\theta}) + \frac{1}{4}(e^{j2\theta} + e^{-j2\theta})$

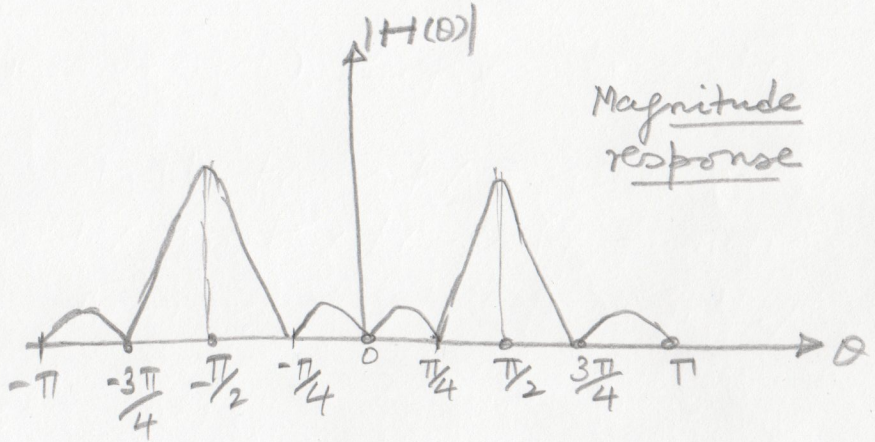
$\frac{Y(\theta)}{X(\theta)} = \frac{3}{2} + 2\cos\theta + \frac{1}{2}\cos 2\theta$ ← Zero phase

↑ Real Part only

(a) $H(z) = \frac{Y(z)}{X(z)} = \frac{1}{8}\left(\frac{1 - z^{-8}}{1 + z^{-2}}\right)$ — 8 zeros; Two poles



Two zeros will be cancelled by TWO poles



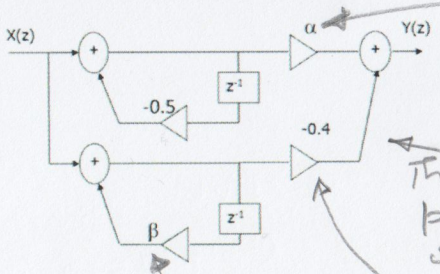
Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

5. The following figure shows an IIR filter structure, and its transfer function given by

$$H(z) = \frac{Y(z)}{X(z)} = \frac{3z(5z-2)}{(z+0.5)(3z-1)}$$

Determine the values of the Multiplier Coefficients α and β.



This is a parallel structure

$$H(z) = \frac{Y(z)}{X(z)} = \frac{3z(5z-2)}{(z+0.5)(3z-1)} = \frac{(5-2z^{-1})}{(1+0.5z^{-1})(1-\frac{1}{3}z^{-1})}$$

$$\frac{5-2z^{-1}}{(1+\frac{1}{2}z^{-1})(1-\frac{1}{3}z^{-1})} = \frac{A}{1+\frac{1}{2}z^{-1}} + \frac{B}{1-\frac{1}{3}z^{-1}}$$

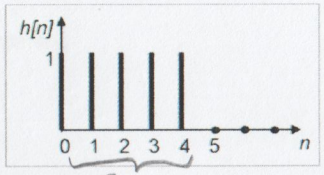
$\beta = \frac{1}{3}$

$\alpha = \frac{27}{5}$

$$A = \frac{5-2(-2)}{1-\frac{1}{3}(-2)} = \frac{27}{5}$$

$$B = \frac{5-2(3)}{1+\frac{1}{2}(3)} = -\frac{2}{5}$$

6. A FIR filter has an impulse response as shown below:



(a) Show that the frequency response of this filter is given by

$$H(\theta) = e^{-j(2\theta)} \frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{\theta}{2}\right)}$$

(b) Draw an implementation of this filter as a cascade structure.

(a) $H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$

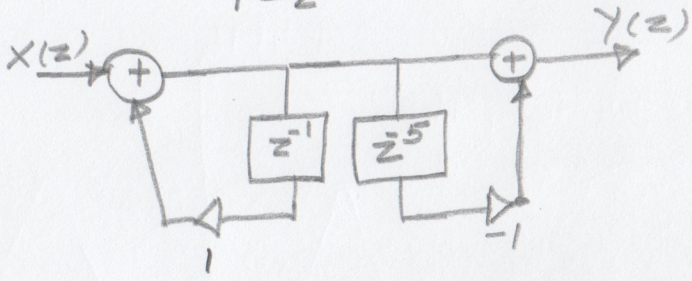
$$H(z) = \frac{1-z^{-5}}{1-z^{-1}}$$

$$H(\theta) = \frac{1-e^{-5\theta j}}{1-e^{-j\theta}} = \frac{e^{-j\frac{5}{2}\theta}}{e^{-j\frac{1}{2}\theta}} \frac{(e^{j\frac{5}{2}\theta} - e^{-j\frac{5}{2}\theta})}{(e^{-j\frac{1}{2}\theta} - e^{-j\frac{1}{2}\theta})}$$

$$H(\theta) = e^{-j2\theta} \frac{\sin\frac{5\theta}{2}}{\sin\frac{\theta}{2}}$$

(b)

$$H(z) = \frac{1-z^{-5}}{1-z^{-1}}$$

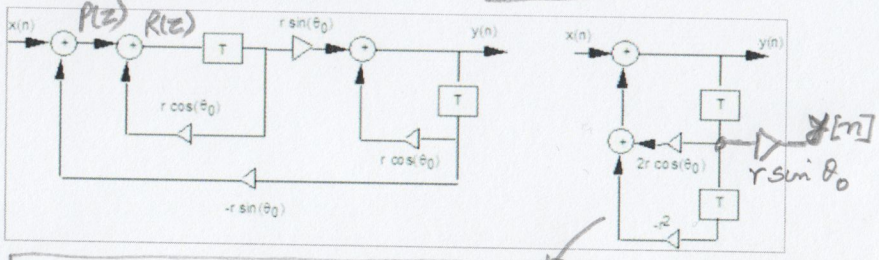


Cascade structure

Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

7. Show that the systems (i) and (ii) shown below are equivalent:



$$R(z) = X(z) + z^{-1}Y(z)(-r\sin\theta_0) + (r\cos\theta_0)R(z)z^{-1}$$

$$R(z)(1 - r\cos\theta_0 z^{-1}) = X(z) - r\cos\theta_0 z^{-1}Y(z) \quad \text{— (1)}$$

$$Y(z) = R(z)(r\sin\theta_0)z^{-1} + Y(z)\,r\cos\theta_0\,z^{-1}$$

$$R(z) = \frac{Y(z)[1 - r\cos\theta_0 z^{-1}]}{r\sin\theta_0 z^{-1}} \quad \text{---- (2)}$$

From (1) & (2)

$$\frac{X(z) - r\sin\theta_0 z^{-1}Y(z)}{1 - r\cos\theta_0 z^{-1}} = \frac{Y(z)[1 - r\cos\theta_0 z^{-1}]}{r\sin\theta_0 z^{-1}}$$

$$\frac{Y(z)}{X(z)} = \frac{r\sin\theta_0 z^{-1}}{1 - 2r\cos\theta_0 z^{-1} + r^2 z^{-2}}$$

Both are equivalent!

$$\therefore \frac{Y(z)}{X(z)} = \frac{r\sin\theta_0 z^{-1}}{1 - 2r\cos\theta_0 z^{-1} + r^2 z^{-2}}$$

8. A third order FIR filter has a transfer function G(z) given by

$$G(z) = (9 + 6z^{-1} - 8z^{-2})(2 + 5z^{-1}).$$

From G(z), determine the transfer function of a FIR filter whose magnitude response is identical to that of G(z) and has

(a) a minimum phase response
(b) a maximum phase response

$$G(z) = (3 - 2z^{-1})(3 + 4z^{-1})(2 + 5z^{-1})$$

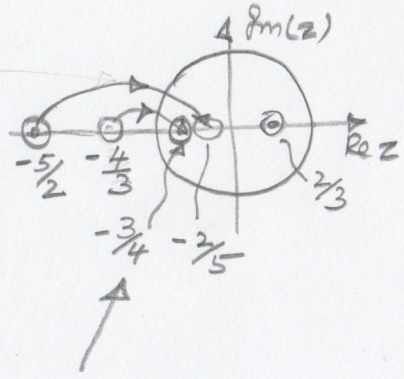
(a) $$G(z) = 18\left(1 - \tfrac{2}{3}z^{-1}\right)\left(1 + \tfrac{4}{3}z^{-1}\right)\left(1 + \tfrac{5}{2}z^{-1}\right)$$

(<1, >1, >1)

$$G(z) = 60\left(1 - \tfrac{2}{3}z^{-1}\right)\left(1 + \tfrac{3}{4}z^{-1}\right)\left(1 + \tfrac{2}{5}z^{-1}\right)$$

(<1, <1, <1)

For minimum phase reflect the zeros inside the unit circle

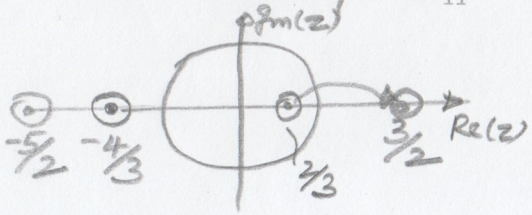


(b) $$G(z) = 12\left(1 - \tfrac{3}{2}z^{-1}\right)\left(1 + \tfrac{4}{3}z^{-1}\right)\left(1 + \tfrac{5}{2}z^{-1}\right)$$

(>1, >1, >1)

all the zeros are outside the unit circle

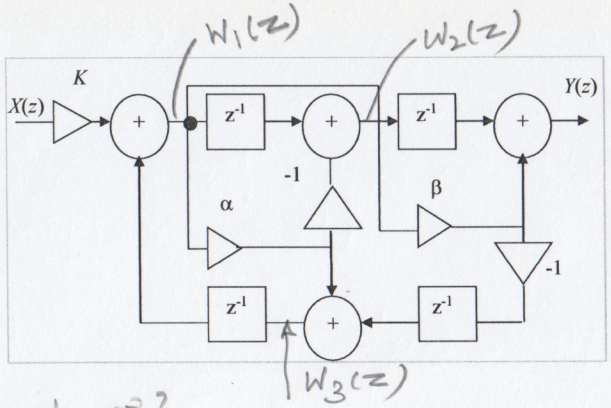
$\therefore$ G(z) has maximum phase response



Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

9. Analyse the digital structure shown on the right:



(a) Determine its transfer function $H(z)$ =(Y(z))/(X(z))

(b) What should be the value of the multiplier coefficient K so that the transfer function *H(z)* that you obtained in (a) above has a unity gain at θ = 0?

(c) What should be the value of the multiplier coefficient *K* so that *H*(*z*) has a unity gain at θ = π?

(d) Is there a difference between these two values of *K* obtained in Question (b) and (c)? If not, why not?

(a) $W_1(z) = K X(z) + W_3(z) z^{-1}$ — ①

$W_2(z) = W_1(z) z^{-1} + (-\alpha) W_1(z)$ — ②

$W_3(z) = \alpha W_1(z) - \beta W_1(z) z^{-1}$ — ③

$Y(z) = W_2(z) z^{-1} + \beta W_1(z)$ — ④

From ① & ③ $W_1(z) = K X(z) + z^{-1}(\alpha - \beta z^{-1}) W_1(z)$

$\therefore W_1(z) = \dfrac{K X(z)}{1 - \alpha z^{-1} + \beta z^{-2}}$

From ② & ④ $Y(z) = [W_1(z) z^{-1} - \alpha W_1(z)] z^{-1} + \beta W_1(z)$

$Y(z) = \{[z^{-1} - \alpha] z^{-1} + \beta\} W_1(z)$

$Y(z) = K X(z) \dfrac{z^2 - \alpha z^{-1} + \beta}{1 - \alpha z^{-1} + \beta z^{-2}}$

$\therefore \dfrac{Y(z)}{X(z)} = \dfrac{K[\beta - \alpha z^{-1} + z^{-2}]}{1 - \alpha z^{-1} + \beta z^{-2}}$

(b) $H(z) = K \dfrac{[\beta - \alpha z^{-1} + z^{-2}]}{1 - \alpha z^{-1} + \beta z^{-2}}$

$\therefore H(\theta)\big|_{\theta=0} = K \dfrac{\beta - \alpha + 1}{1 - \alpha + \beta} \Rightarrow$ $\boxed{K = \dfrac{1 - \alpha + \beta}{\beta - \alpha + 1} = 1}$

($H(\theta)\big|_{\theta=0} = 1$ given)

(c) $H(\theta) = K \cdot \dfrac{\beta - \alpha e^{-j\theta} + e^{-j2\theta}}{1 - \alpha e^{-j\theta} + \beta e^{-j2\theta}}$; $H(\theta)\big|_{\theta=\pi} = 1$ (given)

$1 = K \dfrac{\beta - \alpha e^{-j\pi} + e^{-j2\pi}}{1 - \alpha e^{-j\pi} + \beta e^{-j2\pi}} = K\left(\dfrac{\beta - \alpha + 1}{1 - \alpha + \beta}\right)$

$\therefore \boxed{K = 1}$

(d) $H(0) = 1$

$H(\pi) = 1$

$H(z)$ is an all-pass function with a constant magnitude at all values of θ.

Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

10. (a) Calculate the magnitude and phase responses of the following FIR filter:

$$y[n] = \frac{1}{4}[x[n] - 2x[n-1] + x[n-2]]$$

(b) Sketch the magnitude and phase responses.

(c) Show that the above FIR filter has a constant group delay.

(a) $H(z) = \frac{1}{4}(1 - 2z^{-1} + z^{-2})$

$H(\theta) = \frac{1}{4}(1 - 2e^{-j\theta} + e^{-j2\theta})$

$= \frac{1}{4} e^{-j\theta}(e^{j\theta} - 2 + e^{-j\theta})$

$= \frac{1}{4} e^{-j\theta}(2\cos\theta - 2)$

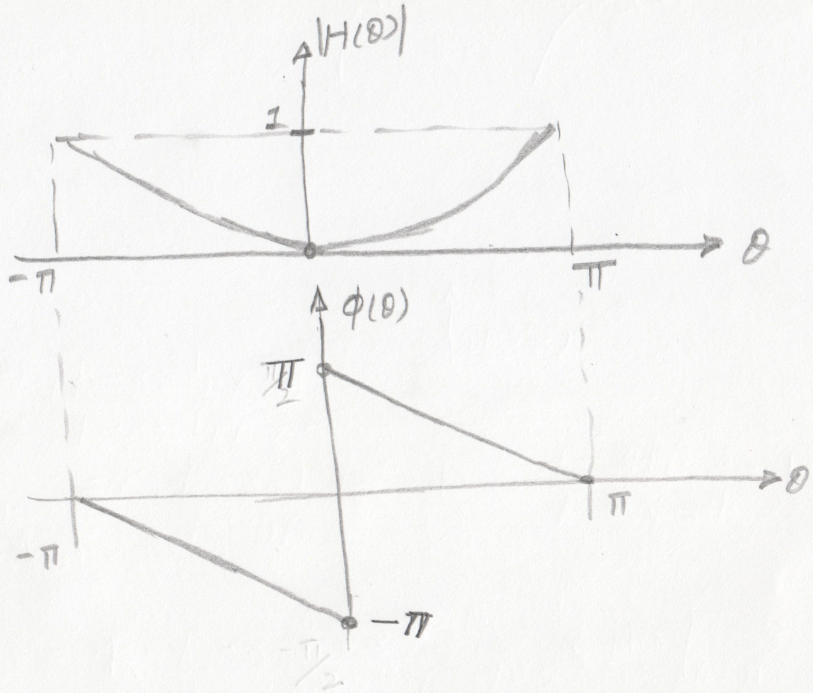
$= e^{-j\theta} \cdot \frac{1}{2}(\cos\theta - 1)$

$B(\theta)$

$|H(\theta)| = |B(\theta)| = \left|\frac{1}{2}(\cos\theta - 1)\right|$ is always negative

$\phi(\theta) = -\theta \pm \pi \quad B(\theta) \leq 0$

(b)



(c) group delay $\tau_g = -\frac{d\phi(\theta)}{d\theta}$

$= -\frac{1}{d\theta}(-\theta \pm \pi)$

$= 1$

Constant

Note: $\int \theta e^{jk\theta} d\theta = \frac{e^{jk\theta}}{k^2}(1-jk\theta)$ ← This will be given for Q1

Topic 3: Tutorial questions (Level 4)

SOLO – Level 4

1. The frequency response of an ideal differentiator is given by

$$H(\theta) = j\theta e^{-j\theta\tau} \quad -\pi \le \theta \le \pi$$

This response is periodic with period 2π. The quantity τ is the delay of the system in samples.

(a) Sketch the magnitude and phase responses of the system for $-\pi \le \theta \le \pi$.

(b) Find the impulse response *h[n]* of the system as a sum of a sine and cosine functions.

$H(\theta) = j\theta e^{-j\theta\tau}$ $-\pi \le \theta \le \pi$

$|H(\theta)| = \theta \quad 0 \le \theta \le \pi$

$= -\theta \quad -\pi < \theta < 0$

Magnitude response

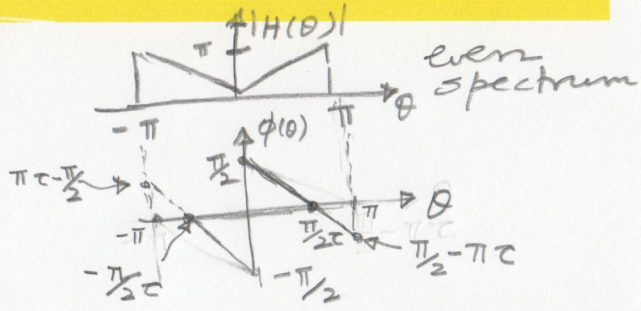
$H(\theta) = e^{j\frac{\pi}{2}} \cdot \theta \cdot e^{-j\theta\tau}$

$= j\, e^{-j(\theta\tau - \pi/2)}$

$H(\theta) = \theta\, e^{\ldots}$

$\therefore \phi(\theta) = -\theta\tau + \pi/2 \quad \theta \ge 0$

$= -\theta\tau - \pi/2 \quad \theta < 0$



(b)

$h[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\theta)\, e^{j\theta n}\, d\theta$

$= \frac{1}{2\pi}\int_{-\pi}^{\pi} j\theta\, e^{-j\theta\tau}\, e^{j\theta n}\, d\theta$

$h[n] = \frac{j}{2\pi}\int_{-\pi}^{\pi} \theta\, e^{j(n-\tau)\theta}\, d\theta$

$h[n] = \frac{j}{2\pi}\left[\frac{e^{j(n-\tau)\theta}}{(n-\tau)^2}(1-j(n-\tau)\theta)\right]_{-\pi}^{\pi}$

$= \frac{j}{2\pi(n-\tau)^2}\left[\left\{e^{j(n-\tau)\pi}(1-j(n-\tau)\pi)\right\} - \left\{e^{j(n-\tau)(-\pi)}(1-j(n-\tau)(-\pi)\right\}\right]$

$= \frac{1}{2\pi(n-\tau)^2}\left[\left\{e^{j(n-\tau)\pi} - e^{-j(n-\tau)\pi}\right\} - \left\{j(n-\tau)\pi\, e^{j(n-\tau)\pi} - j(n-\tau)\pi\, e^{-j(n-\tau)\pi}\right\}\right]$

$h[n] = -\frac{\sin[\pi(n-\tau)]}{\pi(n-\tau)^2} + \frac{\cos[\pi(n-\tau)]}{n-\tau} \quad n \ne \tau$

2. Determine the magnitude and phase response of the multipath channel *y[n] = x[n] + x[n-M]*. At what frequencies $|H(\theta)| = 0$?

$y[n] = x[n] + x[n-M] \;\therefore\; Y(z) = X(z) + X(z)z^{-M}$

$H(z) = 1 + z^{-M} \Rightarrow H(\theta) = 1 + e^{-j\theta M} = (1+\cos M\theta) + j\sin(M\theta)$

$|H(\theta)| = \sqrt{(1+\cos M\theta)^2 + (\sin M\theta)^2} = \sqrt{2+2\cos(M\theta)}$

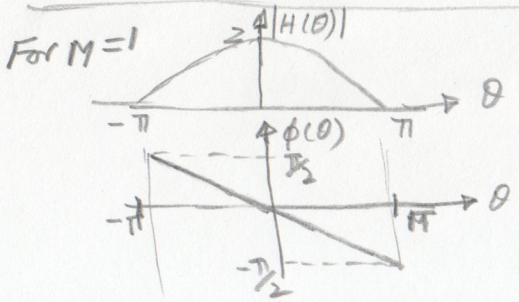
$\phi(\theta) = \tan^{-1}\left\{\frac{-\sin M\theta}{1+\cos M\theta}\right\}$; let $M = 1$

$\phi(\theta) = \tan^{-1}\frac{-\sin\theta}{1+\cos\theta}$

$= \tan^{-1}\left\{\frac{-2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2+2\cos^2\frac{\theta}{2}-1}\right\}$

$\phi(\theta) = -\theta/2$

For M=1



$|H(\theta)| = 0 \Rightarrow \cos M\theta = -1$

$M\theta = \pi$

$\therefore \theta = \pi/M$

Topic 3: Tutorial questions (Level 4)

SOLO – Level 4

3. Let a causal linear time invariant discrete-time system be characterised by a real impulse response $h[n]$ with a DTFT of $H(\theta)$. Consider the system shown in Figure A, where $x[n]$ is a finite length sequence. Determine the frequency response of the overall system $G(\theta)$ $\{G(z) = {}^{Y(z)}/_{X(z)}\}$ in terms of $H(\theta)$ and show that it has a zero-phase response.

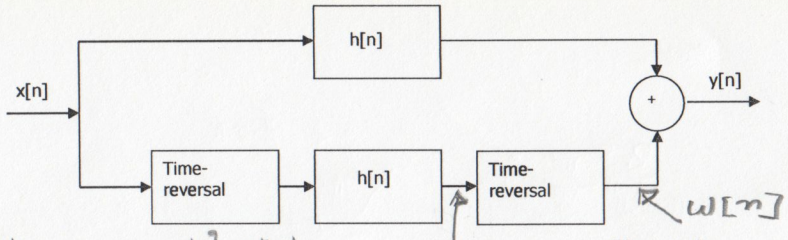


Figure A

$h[n] \xrightarrow{DTFT} H(\theta)$

$v[n] = x[-n] * h[n]$

$w[n] = v[-n] = x[n] * h[n]$ $\quad = w[n]$

$y[n] = h[n] * x[n] + x[n] * h[-n]$

$y[n] = \{h[n] + h[-n]\} * x[n]$

$Y(z) = [H(z) + H(z^{-1})] X(z)$

$\therefore G(z) = \frac{Y(z)}{X(z)} = H(z) + H(z^{-1})$

$G(\theta) = H(\theta) + H^*(\theta)$

$= (a + jb) + (a - jb) = 2a$

$G(\theta)$ is real

$\therefore G(\theta)$ has zero phase.

4. An analogue system may be implemented as shown in Figure B. Using the transformation $\frac{1}{s} \rightarrow k\frac{1+z^{-1}}{1-z^{-1}}$ a discrete time system may be obtained; where k is a constant.

   (a) Draw the digital structure of the system, using the minimum number of multipliers and delay units.

   (b) Write the difference equation that governs the behaviour of the system in the form $y[n] = 2\cos[\theta_0]\, y[n-1] - y[n-2]$ and find the value of $\cos\theta_0$.

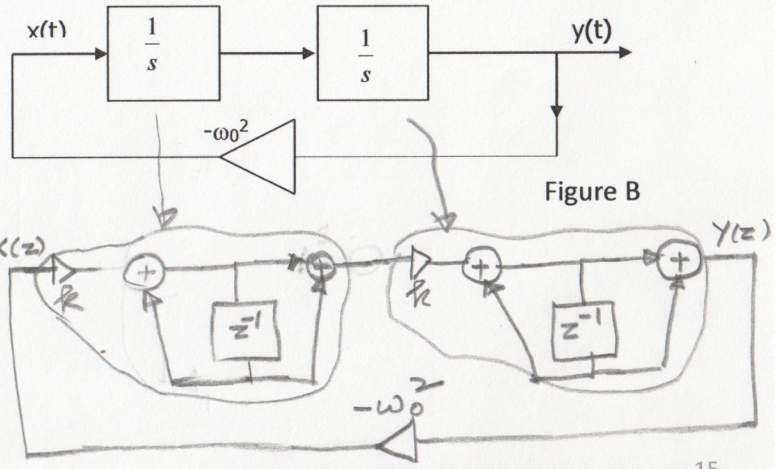


Figure B

(a)

Number of delays = 2

(b) $\frac{Y(z)}{X(z)} = k\frac{1+z^{-1}}{1-z^{-1}} \cdot k\frac{1+z^{-1}}{1-z^{-1}} = k^2\frac{1+2z^{-1}+z^{-2}}{1-2z^{-1}+z^{-2}}$ $\quad -①$

$X(z) = -\omega_0^2 Y(z) \Rightarrow x[n] = -\omega_0^2 y[n]$ $\quad -②$

$y[n] = k^2[x[n] + 2x[n-1] + x[n-2]] + 2y[n-1] - y[n-2]$

$y[n] = k^2\{-\omega_0^2 y[n] - 2\omega_0^2 y[n-1] - \omega_0^2 y[n-2]\} + 2y[n-1] - y[n-2]$

$y[n]\{1 + k^2\omega_0^2\} = 2(1 - k^2\omega_0^2)\, y[n-1] - (1 + k^2\omega_0^2)\, y[n-2]$

$\therefore y[n] = 2\frac{(1 - k^2\omega_0^2)}{(1 + k^2\omega_0^2)} y[n-1] - y[n-2] \Rightarrow$ $\frac{1 - k^2\omega_0^2}{1 + k^2\omega_0^2} = \cos\theta_0$

$y[n] = 2\cos\theta_0\, y[n-1] - y[n-2]$

$\left[\frac{d}{dx}\tan^{-1}u = \frac{1}{1+u^2}\frac{du}{dx}\right]$

Topic 3: Tutorial questions (Level 5)

SOLO – Level 5

1. (a) Find the group delay for each of the following system, where $|\alpha| < 1$

(i) $H_1(\theta) = 1 - \alpha e^{-j\theta}$ (ii) $H_2(\theta) = \frac{1}{1-\alpha e^{-j\theta}}$

(b) using the results obtained in (a) above, show that the group delay of an all-pass filter with a transfer function $H_0(z) = \frac{z^{-1}-\alpha}{1-\alpha z^{-1}}$ is $\frac{1-\alpha^2}{1+\alpha^2-2\alpha\cos(\theta)}$

(i) $H_1(\theta) = 1-\alpha e^{-j\theta}$; $H(\theta) = (1-\alpha\cos\theta) + j\alpha\sin\theta$

$\phi_1(\theta) = \tan^{-1}\frac{\alpha\sin\theta}{1-\alpha\cos\theta}$

The group delay $\tau_1(\theta) = -\frac{d}{d\theta}\phi_1(\theta) = -\frac{1}{1+\left(\frac{\alpha\sin\theta}{1-\alpha\cos\theta}\right)^2}\frac{d}{d\theta}\left(\frac{\alpha\sin\theta}{1-\alpha\cos\theta}\right)$

$\tau_1(\theta) = \frac{-1}{1+\left(\frac{\alpha\sin\theta}{1-\alpha\cos\theta}\right)^2}\cdot\frac{(1-\alpha\cos\theta)\alpha\cos\theta - (\alpha\sin\theta)^2}{(1-\alpha\cos\theta)^2}$

which after simplification

$$\tau_1(\theta) = \frac{\alpha^2-\alpha\cos\theta}{1+\alpha^2-2\alpha\cos\theta}$$

$H_0(z) = \frac{-\alpha\left[1-\frac{1}{\alpha}z^{-1}\right]}{1-\alpha z^{-1}} = -\alpha\cdot\underbrace{\left(1-\frac{1}{\alpha}z^{-1}\right)}_{\text{Part (i) above}}\cdot\underbrace{\frac{1}{1-\alpha z^{-1}}}_{\text{Part (ii) above}}$

The group delay of $H_0(z)$ is $\tau_3(\theta) = -\alpha\cdot\frac{\left(\frac{1}{\alpha}\right)^2-\frac{1}{\alpha}\cos\theta}{1+\left(\frac{1}{\alpha}\right)^2-2\left(\frac{1}{\alpha}\right)\cos\theta} + -\frac{(\alpha^2-\alpha\cos\theta)}{1+\alpha^2-2\alpha\cos\theta}$

$\leftarrow$ simplify

$$\tau_3(\theta) = \frac{1-\alpha^2}{1+\alpha^2-2\alpha\cos\theta}$$

(ii) Having found the group delay for $H_1(\theta) = 1-\alpha e^{-j\theta}$, the group delay for $H_2(\theta)$ is the inverse of $H_1(\theta)$

$\therefore \phi_2(\theta) = -\phi_1(\theta)$

$\therefore \tau_2(\theta) = -\tau_1(\theta) = -\frac{(\alpha^2-\alpha\cos\theta)}{1+\alpha^2-2\alpha\cos\theta}$

2. Find the minimum phase system that has a magnitude response given by

$$|H(\theta)|^2 = \frac{\frac{5}{4}-\cos\theta}{\frac{10}{9}-\frac{2}{3}\cos\theta}$$

To solve this problem consider $|H(\theta)|^2$ in terms of complex exponentials:

$$|H(\theta)|^2 = \frac{\frac{5}{4}-\frac{1}{2}e^{j\theta}-\frac{1}{2}e^{-j\theta}}{\frac{10}{9}-\frac{1}{3}e^{j\theta}-\frac{1}{3}e^{-j\theta}} \quad \left\{\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}\right\}$$

Replacing $e^{j\theta}$ by z and $e^{-j\theta}$ by z^{-1}, this becomes

$$H(z)H(z^{-1}) = \frac{\frac{5}{4}-\frac{1}{2}z-\frac{1}{2}z^{-1}}{\frac{10}{9}-\frac{1}{3}z-\frac{1}{3}z^{-1}} = \frac{\left(1-\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{2}z\right)}{\left(1-\frac{1}{3}z^{-1}\right)\left(1-\frac{1}{3}z\right)}$$

The minimum phase system is then formed by extracting the poles and zeros that are inside the unit circle

$$H_{min}(z) = \frac{1-\frac{1}{2}z^{-1}}{1-\frac{1}{3}z^{-1}}$$

Topic 3: Tutorial questions (Level 5)

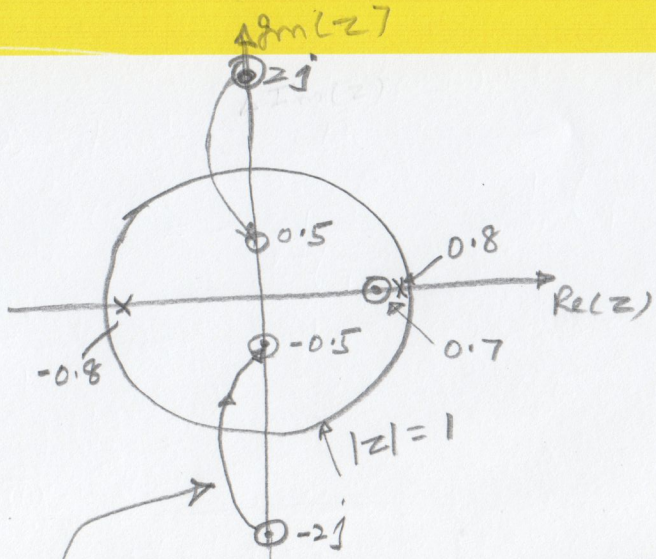
SOLO – Level 5

3. A stable and causal system (H(z)) may always be factored into a product of a minimum phase system with an all-pass system: $H(z) = H_{min}(z) \cdot H_{ap}(z)$ **or** $H(z) = H_{min}(z) \cdot H_{lp}(z)$; where $H_{lp}(z)$ is a linear phase system.

$$H(z) = \frac{(1 - 0.7z^{-1})(1 - j2z^{-1})(1 + j2z^{-1})}{(1 - 0.8z^{-1})(1 + 0.8z^{-1})}$$

(a) Find a minimum phase system function $H_{min}(z)$ and an all-pass system $H_{ap}(z)$, for the above H(z).

(b) Find a minimum phase system function $H_{min}(z)$ and a linear phase system $H_{lp}(z)$, for the above H(z).



(a) $H(z) = H_{min}(z) \cdot H_{ap}(z)$

The H(z) has non minimum phase factors $(1 - j2z^{-1})(1 + j2z^{-1})$ which may be written as the product of a minimum phase term and all-pass factors:

$$H(z) = \underbrace{\frac{(1 - 0.7z^{-1})\ (1 - \frac{1}{2}jz^{-1})(1 + \frac{1}{2}jz^{-1})}{(1 - 0.8z^{-1})(1 + 0.8z^{-1})}}_{H_{min}} \times \underbrace{\frac{(1 - j2z^{-1})(1 + j2z^{-1})}{(1 - \frac{1}{2}jz^{-1})(1 + \frac{1}{2}jz^{-1})}}_{H_{ap}}$$

Reflect the zeros outside the unit circle to inside the unit circle to obtain min phase filter.

$H_{LP} = (1 - \frac{1}{2}jz^{-1})(1 + \frac{1}{2}jz^{-1})(1 - 2jz^{-1})(1 + 2jz^{-1})$

(b) $H(z) = H_{min}(z) \cdot H_{lp}(z)$

The system is not minimum phase because the factors $(1 - j2z^{-1})\ (1 + j2z^{-1})$ are out side the unit circle. First we reflect the zeros about the unit circle and replace it with poles: $\frac{1}{(1 - \frac{1}{2}jz^{-1})}$ & $\frac{1}{1 + \frac{1}{2}jz^{-1}}$. Then we multiply each term with a linear phase factor that has zeros at $(z = \frac{1}{2}j$ & $z = 2j)$ and $z = -\frac{1}{2}j$ & $z = -2j$

$$\therefore H(z) = \underbrace{\frac{(1 - 0.7z^{-1})}{(1 - 0.8z^{-1})(1 + 0.8z^{-1})} \cdot \frac{1}{(1 - \frac{1}{2}jz^{-1})(1 + \frac{1}{2}jz^{-1})}}_{H_{min}} \times H_{lp}$$

Topic 3: Tutorial questions (Level 5)

SOLO – Level 5

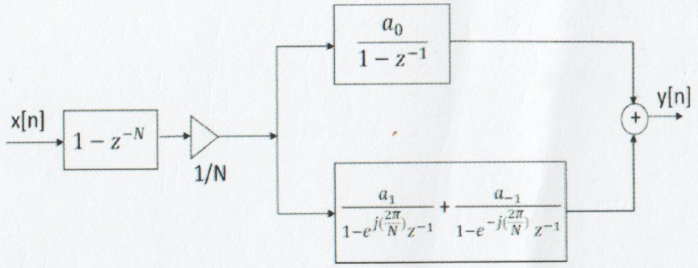
4. A filter structure with a transfer function $\{H(z) = \frac{Y(z)}{X(z)}\}$ is shown below. You may assume that $N = 3$, $a_0 = 3$ and $H(\theta)|_{\theta=2\pi/3} = 6+j3\sqrt{3}$.

   Also $H(e^{j\frac{2\pi k}{N}}) = a_k$ and $H(e^{-j\frac{2\pi k}{N}}) = a_{-k}$

   (a) Determine a_1 & a_{-1} such that this filter has a real valued impulse response *h[n];*

   (b) Find H(z) and draw a filter structure using delays elements, multipliers and adders.

   (c) Show that this filter structure can be realised as an FIR filter.



(a) $a_1 = 6+j3\sqrt{3}$ $\quad N=3$

$a_{-1} = 6-j3\sqrt{3}$

$$H(z) = \frac{1-z^{-3}}{3}\left[\frac{3}{1-z^{-1}} + \frac{6+j3\sqrt{3}}{1-e^{j\frac{2\pi}{3}}z^{-1}} + \frac{6-j3\sqrt{3}}{1-e^{-j\frac{2\pi}{3}}z^{-1}}\right]$$

$$= \frac{1-z^{-3}}{3}\left[\frac{3}{1-z^{-1}} + \frac{(6+j3\sqrt{3})(1-z^{-1}e^{-j\frac{2\pi}{3}}) + (6-j3\sqrt{3})(1-z^{-1}e^{-j\frac{2\pi}{3}})}{1+z^{-1}+z^{-2}}\right]$$

$$= \frac{1-z^{-3}}{3}\left[\frac{3}{1-z^{-1}} + \frac{3(4-z^{-1})}{1+z^{-1}+z^{-2}}\right]$$

$$H(z) = \frac{1-z^{-3}}{3} \cdot 3\left[\frac{5-4z^{-1}+2z^{-2}}{(1-z^{-1})(1+z^{-1}+z^{-2})}\right]$$

$$H(z) = 5-4z^{-1}+2z^{-2}$$

FIR filter.

(b)

